CASCADES26 Expedition

CGGS Amundsen

FMeasurement scenarios for Chipiron

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1 Conventions & definitions of key quantities

1.1 System sketch and quantities: frazil rheology

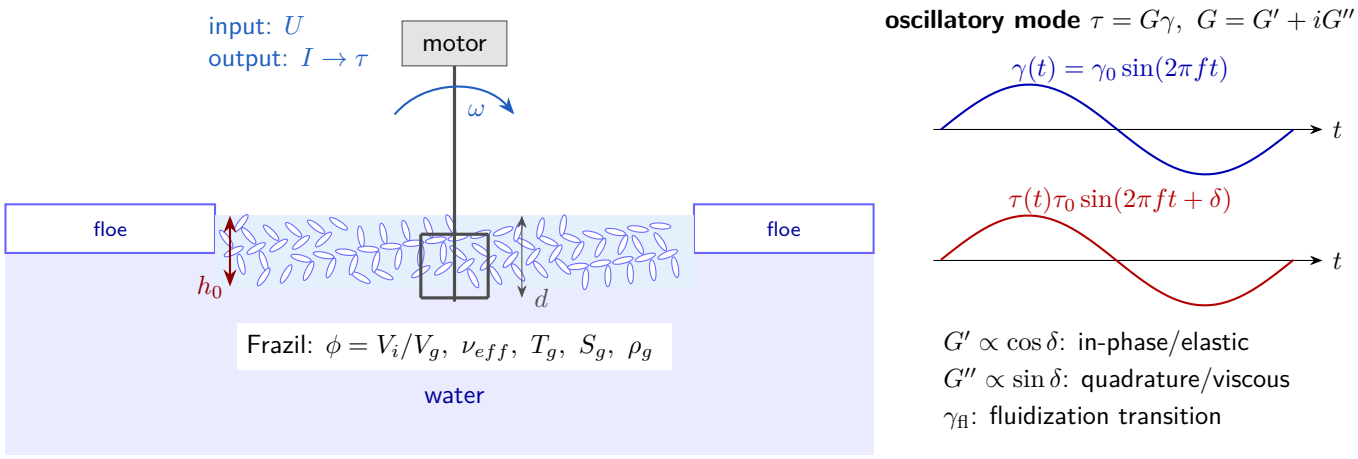


Figure 1. Chipiron in a frazil layer between floes. Continuous mode: imposed ω, measured τ. Oscillatory mode: imposed $\gamma(t)$, measured $\tau(t)$.

Symbol	Meaning	Obtained from
Imposed / measured by the rheometer		
U	motor input voltage	imposed (PC ESC)
ω	rotation speed (continuous mode)	imposed via U; check with calibration (§3)
$I,\ \tau$	motor current, torque	measured (ARD), $\tau \propto I$
$\gamma_0,\ f$	forcing strain amplitude/frequency (oscillatory mode)	imposed via U
$G',\ G''$	storage / loss modulus (oscillatory mode)	from $\tau(t)$ vs. $\gamma(t)$
δ	phase shift between γ and τ	measured
γ_{fl}	fluidization	from $G'(\gamma_0), G''(\gamma_0)$
ν_{eff}	effective viscosity	from viscous response
J	inertia number	from $\tau(\omega)$
Frazil state		
$\phi = V_i/V_g$	solid fraction	SMP FIL GRAD BAL; [3]
h_0	frazil layer thickness	field estimate
$V_g,\ m_g$	sample volume, mass	GRAD, BAL
$T_g,\ S_g$	sample temperature, salinity	THM, CND
d	vane immersion depth	geometry

Table 1. Quantities used throughout for ocntinuous ad oscillatory rheology experiments.

1.2 Equipment symbols and colour code

Colour = what the item is used for. The symbols are used everywhere in the rest of the document.

Symbol	Item	Purpose	Field	Lab
Blue — Inputs/outputs				
PC	laptop	runs everything below	✓	✓
ARD	Arduino / current measurement	acquires I and U	✓	✓
ESC	ESCON Studio	motor control / imposes U	✓	✓
PY	Python scripts	acquisition and processing	✓	✓
Grey — mechanics				
CHIP	Chipiron (rheometer)	measures $\tau(\omega)$, G', G''	✓	✓
FLT	floating pieces	keeps Chipiron at the right depth d	✓	—
Green — sample characterization (ϕ, density, volume, mass)				
SMP	frazil sampler	get sample and possibly filter it	✓	✓
FIL	filter / mesh	crystal–brine separation	✓	✓
GRAD	graduated volume container	V_g, V_i	✓	✓
BAL	balance	m_g	✓	✓
FZM	frazilometers	local frazil thickness	✓	—
CONT	sample containers	for lab rheology and/or characterization	✓	✓
Orange — environment				
THM	thermometer	T (air, water, sample)	✓	✓
CND	conductimeter / salinity	S (if available)	(✓)	✓
Purple — visual record				
CAM	phone / camera	before / during / after pictures	✓	✓
MIC	optics/ camera	crystal size, aspect ratio	(optional)	(optional)

Table 2. Equipment legend.

1.3 Scripts

Purpose	Script	Command / parameters
Plan ω ramp (log / lin)	`Ramp_planning_Chipiron.py`	[ ]
Plan γ_0 sweep (log / lin)	`Oscill_sweep_planning_Chipiron.py`	[ ]
Execute a run	`run.py`	[ ]
Analysis / quick check	[ ]	[ ]
Sample characterization	[ ]	[ ]

Table 3. Scripts (PY). Empty = not decided / not known.

2 CASCADES pipeline

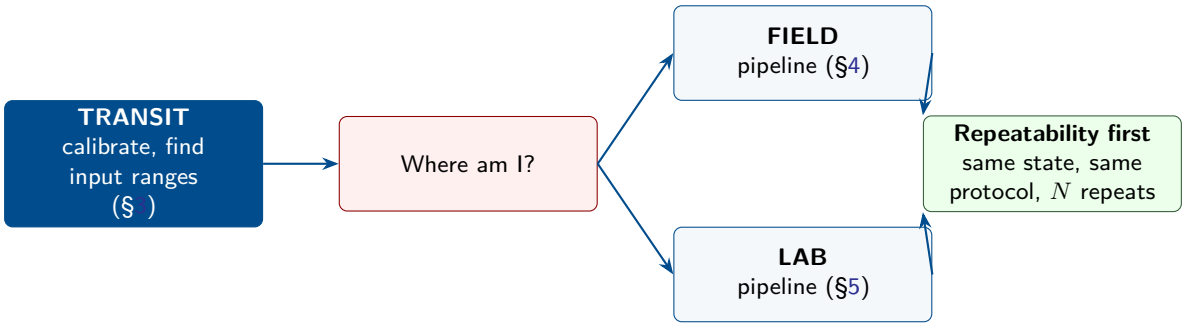


Figure 2. Overall steps. Note: focus on repeatabilit yto make sure we have robust data. Parameter variations only come afterwards. (§**??**).

3 Transit period

Frazil can already be sampled and the setup tested during transit. Goal: calibrate quantities, identify usable ranges, set parameters for all tests.

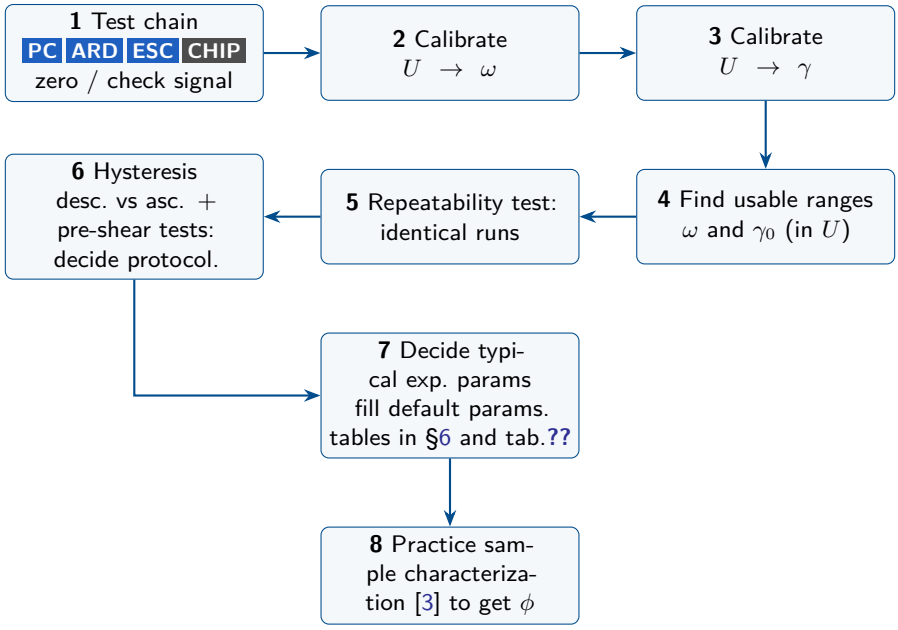


Figure 3. Transit pipeline.

Quantity to find	**Range**	**N pts**	**Time / pt**	**Default**
ω (ramp)	[ ]	[ ]	[ ]	[ ]
γ_0 (oscill.)	[ ]	[ ]	[ ]	[ ]
f (oscill.)	[ ]	[ ]	[ ]	[ ]

Table 4. Experimental ranges to test.

4 Field case

4.1 Pipeline

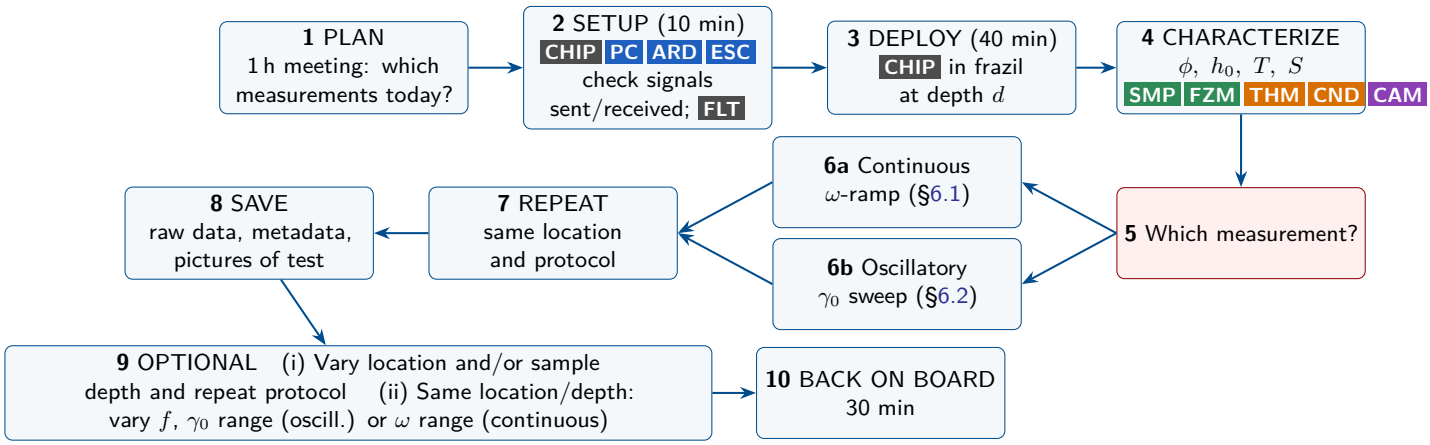


Figure 4. Field pipeline.

Role (3 people)	Actions
Deck handler	sends inputs, runs PC ESC, records run ID + metadata, checks live I/τ (and waveform quality in oscillatory mode)
Operators 1 & 2	position CHIP in the frazil layer, keep geometry / depth d, CAM local frazil, record ice type, h_0, location, wave state, wind, obtain ϕ

Table 5. Field roles.

4.2 Outputs

1. **Context**: date/time, GPS, floe / water configuration, waves, wind, T_{air}, T_{water}, S
2. **Frazil**: ice type, ϕ, h_0, T, crystal size / shape
3. **Geometry**: vane type, d, position vs. floe edge, sample depth
4. **Inputs**: all rows of the protocol table (§6)
5. **Raw data**: full I/τ and U time series (PC ARD)
6. **Pictures / video**: before, during, after (CAM)
7. **Sample history:** previous tests if any, pre-shear if any, visible qualitative structural change
8. **Side sample** from same place and depth: ϕ, T, S, m, V, crystal size (SMP FIL GRAD BAL)

Priority order	
1	one good descending ω ramp + full raw series + local ϕ, h_0 + context
2	repeat same ramp (repeatability / history)
3	one oscillatory amplitude sweep
4	repeat same oscillatory sweep
5	Other location/sampling depth, record changes in ϕ, h_0
6	more f values (oscillatory)

5 Lab case

5.1 Pipeline

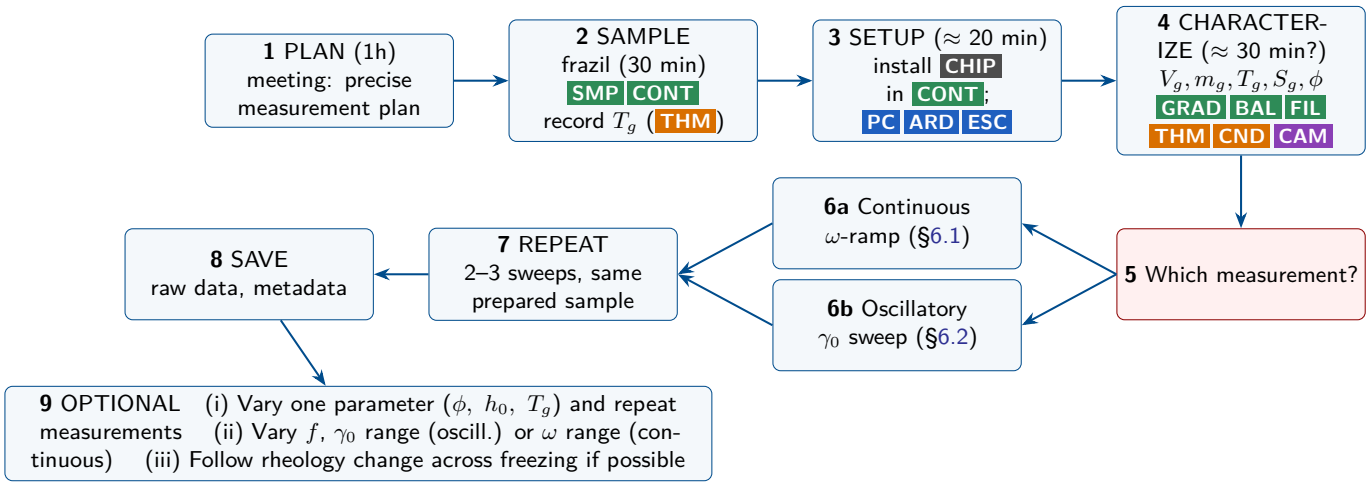


Figure 5. Lab pipeline.

5.2 Roles

Role (2 people)	Actions
Operator 1	controls CHIP / acquisition (PC ARD ESC), checks live outputs, records run IDs, exact protocol, sample history
Operator 2	prepares and characterizes the sample (GRAD BAL FIL THM CND), follows temperature / ice state / structural change

5.3 Outputs

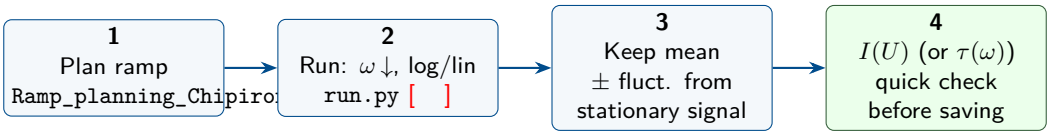
1. **Context**: date/time, GPS, sampling context (depth, waves, wind), T_{room}, T_{sample}, S
2. **Frazil**: ice type, ϕ, h_0, T, crystal size / shape, V_g, T_g, S_g, ρ_g
3. **Geometry**: vane / rotor geometry, d, container dimensions, distance to walls / bottom, corrections / calibration used
4. **Inputs**: all rows of the protocol table (§6)
5. **Raw data**: full I/τ and U time series (PC ARD)
6. **Pictures / video**: before, during, after (CAM)
7. **Sample history:** previous tests if any, pre-shear if any, visible qualitative structural change

Priority order	
1	one descending ω ramp + full raw series + ϕ, h_0 + sample context
2	repeat the ramp (repeatability / history)
3	one oscillatory amplitude sweep
4	repeat oscillatory exp.
5	vary available parameters (ϕ, h_0, T_g, time after sampling)
6	vary sweep amplitudes/frequencies (oscill.)
7	perform frequency (not amplitude) sweep for oscill. measurements ?
8	follow rheology across freezing if controlled conditions allow it

6 Measurement protocols (field and lab)

"Range / N pts / time per pt" refer to the swept quantity; script commands are in Table 3.

6.1 A — Continuous: descending ω ramp



Parameter	Range	N pts	Time / pt	Default field	Default lab
ω ramp (descending)	[ ]	[ ]	[ ]	[ ]	[ ]
Transient time	—	—	[ ]	[ ]	[ ]
Steady averaging window	—	—	[ ]	[ ]	[ ]
Acquisition rate	[ ]	—	—	[ ]	[ ]
Repeats of full ramp	—	[ ]	[ ]	[ ]	[ ]

Table 6. Ramp parameters.

Record	U, ω, I/τ, t continuously over the whole ramp (full $I(t)$, $\tau(t)$ for each ω)
Look for	steady value at each ω; low-speed behaviour: yield-stress, intermittent motion/ arrest/strong fluctuations
Output	$\tau(\omega)$ with uncertainty / fluctuations at each point; repeatability; yield stress
Quick check	live τ (or $I(U)$) vs. ω: curve should increase and plateau (Fig. 6)
Later	fit Herschel–Bulkley; dimensionless representation using inertia number J; compare results with plastic beads tests (PMMH)

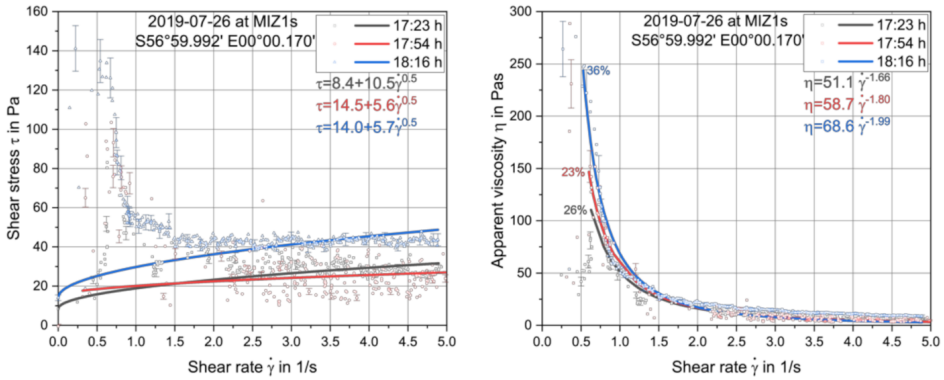
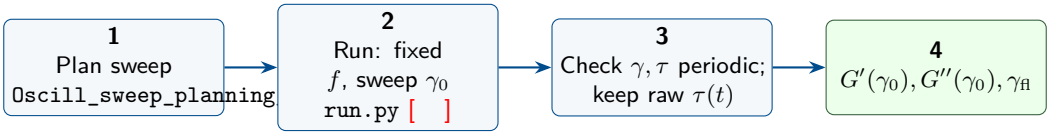


Figure 6. Expected flow curves in frazil, from [4].

Foulques has tested asc/descending rotational speed rampsfrom $1-70$ **rpm**, using **10s** per point (system = plastic beads, Anton Paars rheometer).

6.2 B — Oscillatory: strain-amplitude sweep



Parameter	Range	N pts	Time / pt	Default field	Default lab
γ_0 sweep (log / lin)	[ ]	[ ]	[ ]	[ ]	[ ]
Cycles per point	—	[ ]	—	[ ]	[ ]
Forcing frequency f (fixed)	[ ]	—	—	[ ]	[ ]
Settling time per point	—	—	[ ]	[ ]	[ ]
Sampling rate	[ ]	—	—	[ ]	[ ]
Repeats of full sweep	—	[ ]	[ ]	[ ]	[ ]
f sweep in linear regime (lab, $G'(f), G''(f)$)	[ ]	[ ]	[ ]	—	[ ]

Table 7. Oscillatory parameters.

Record	imposed $\gamma(t)$ and $U(t)$, f, $\tau(t)$ or $I(t)$, number of cycles, T
Derive	G', G'', δ, ν_{eff}; keep raw $\tau(t)$ and harmonic content if non-sinusoidal (nonlinear / fluidised behaviour)
Look for	low-strain linear plateau, departure from linearity, change / crossing of G' and G'', γ_{fl}
Output	$G'(\gamma_0)$, $G''(\gamma_0)$, δ, raw $\tau(t)$, γ_{fl} where identifiable (lab: also G', G'' vs. f ?)
Quick check	input and torque periodic and phase-resolved; qualitatively different regimes (Fig. 7); repeatability

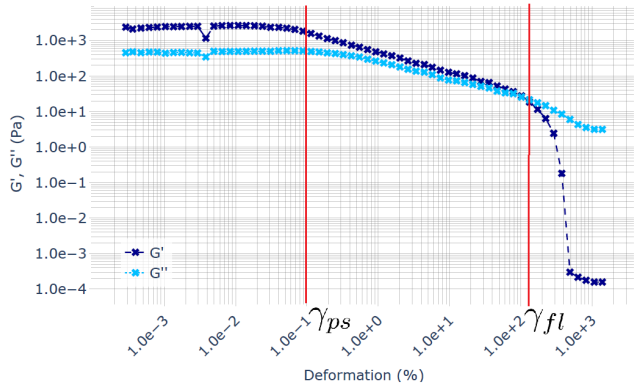


Figure 7. $G'(\gamma_0)$, $G''(\gamma_0)$ in plastic beads (F. Pillet).

Foulques has tested deformation sweeps $\gamma_0 = 10^{-4} - 10^{3}\%$ at $2\pi f = 0.5,\ 5,\ 10,\ 15,\ 20,\ 25$ rad/s, using **5s** per point, **60** points and **300s** intervals. (system = plastic beads, Anton Paar rheometer). He also tried frequency sweeps: $2\pi f = 10^{-1} - 10^{2}$ **rad/s** at $\gamma_0 = 1\%$, and $2\pi f = 10^{-2} - 10^{2}$ **rad/s** at $\gamma_0 = 0.005\%$. One sweep takes about 5min for him.

7 To-do and open questions

☐	**Item**
☐	Number of points per sweep, time per point, range, measurement-time estimate: field ramp, field oscillatory, lab ramp, lab oscillatory (tables in §6)
☐	Lab ramp: active measurement time (check Foulques' data)
☐	Oscillatory (field and lab): time per amplitude sweep and per frequency sweep (check Foulques' notes)
☐	Number of repeats: field ramp (3–4?), lab ramp (2–3?), amplitude sweeps (?)
☐	Total time estimates per scenario; propose order of experiments with concrete times
☐	Emphasis on repeatability vs. parameter change in field and in lab $>$ do we agree on that?
☐	Real command format for `run.py` (ramp and oscillatory); analysis and sample-characterisation scripts
☐	Frazil state (ϕ, h_0): measure just before or just after the run?
☐	Lab oscillatory: do we do a f sweep in the linear regime ?
☐	Calibration methods for $U \to \omega$ and $U \to \gamma$ (fill transit table)
☐	Check order of steps!! eg: do we characterize samples before/after measurements?
☐	Check Foulques' notes: how long is a typical experiment? Typical ranges?
☐	IMPORTANT: for fiels measurements, we'll probably be in cases where waves/wind introducs oscillatory components to measurements and change ϕ,h_0 as we measure. So we can keep it simple and measure only using speed ramps at first order - this won't be that easy to understand.

References

[1] S. De la Rosa and S. Maus, "Laboratory study of frazil ice accumulation under wave conditions," *The Cryosphere*, vol. 6, pp. 173–191, 2012. doi:10.5194/tc-6-173-2012.

[2] S. De la Rosa, S. Maus, and S. Kern, "Thermodynamic investigation of an evolving grease to pancake ice field," *Annals of Glaciology*, vol. 52, no. 57, pp. 206–214, 2011. doi:10.3189/172756411795931787.

[3] S. Maus and S. De La Rosa, "Salinity and solid fraction of frazil and grease ice," *Journal of Glaciology*, vol. 58, no. 208, 2012. doi:10.3189/2012JoG11J110.

[4] F. Paul, T. Mielke, C. Schwarz, J. Schröder, T. Rampai, S. Skatulla, R. R. Audh, E. Hepworth, M. Vichi, and D. C. Lupascu, "Frazil Ice in the Antarctic Marginal Ice Zone," *Journal of Marine Science and Engineering*, vol. 9, no. 6, p. 647, 2021. doi:10.3390/jmse9060647.